

GED Science Practice Test 11

PART A — QUESTIONS

GED-SCI11-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies how sleep interruption affects memory in adult volunteers. Group A sleeps for 8 hours without interruption. Group B is awakened briefly every 90 minutes but still spends 8 hours in bed. The next morning, both groups complete the same word-recall test.

Group | Average words recalled out of 30

A | 24

B | 17

Which conclusion is best supported by the data?

- A) Sleep interruption was associated with lower recall scores.
- B) Sleep interruption increased recall scores.
- C) Both groups had the same average recall score.
- D) Memory cannot be measured with a recall test.

GED-SCI11-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

Potential energy can be calculated as $\text{weight} \times \text{height}$.

Object | Weight (N) | Height above floor (m)

A | 20 | 4

B | 35 | 2

C | 12 | 8

D | 50 | 1

Which object has the greatest gravitational potential energy?

- A) Object A
- B) Object B
- C) Object C
- D) Object D

GED-SCI11-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A region has large deposits of copper ore. Mining copper can provide material for electrical wiring

and electronics, but it can also disturb land, use water, and produce waste rock. Recycling copper from old wiring can reduce the need for new mining.

Which statement best describes one benefit of recycling copper?

- A) It can reduce the amount of new copper ore that must be mined.
- B) It creates unlimited copper atoms from sunlight.
- C) It prevents all minerals from forming naturally.
- D) It makes copper unnecessary in electrical devices.

GED-SCI11-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The endocrine system helps maintain blood glucose levels. When blood glucose rises after a meal, the pancreas releases insulin. Insulin helps cells take in glucose and helps the liver store glucose as glycogen. When blood glucose falls, the pancreas releases glucagon, which signals the liver to release glucose.

Which idea is best illustrated by this process?

- A) Hormones help regulate internal conditions through feedback.
- B) Hormones prevent cells from using glucose.
- C) The pancreas stores all oxygen for the body.
- D) Blood glucose never changes after meals.

GED-SCI11-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A 40 N force moves a crate 6 meters across a smooth floor.

Work = force $\times$ distance

How much work is done on the crate in joules?

Type your answer as a whole number.

GED-SCI11-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how the length of a lever arm affects the input force needed to lift a 100 N load. The student changes the distance from the fulcrum to where the input force is applied. The same load and same lever are used in each trial.

Which TWO factors should be kept constant?

- A) Weight of the load
- B) Type of lever used
- C) Length of the input lever arm
- D) Input force needed
- E) Final measured value

GED-SCI11-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A child inherits one allele for normal hemoglobin and one allele for sickle-cell hemoglobin. Both forms of hemoglobin can be present in the blood. This inheritance pattern is an example of [D1], and the alleles affect the structure of a [D2].

Dropdowns:

D1) codominance | erosion | condensation

D2) protein | planet | mineral

GED-SCI11-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student tests an unknown liquid with universal indicator paper. The paper turns orange-red, which the indicator chart shows as pH 4. The student then adds the liquid to a small amount of baking soda, and bubbles form.

Which conclusion is best supported?

- A) The liquid is acidic.
- B) The liquid is strongly basic.
- C) The liquid is pure oxygen gas.
- D) The liquid is a metal.

GED-SCI11-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A climate scientist compares two gases. Nitrogen makes up most of Earth's atmosphere but absorbs very little infrared radiation. Methane is present in much smaller amounts but absorbs infrared radiation strongly. Infrared absorption by gases can trap heat in the atmosphere.

Which statement is best supported by this information?

- A) Methane can influence atmospheric warming even at lower concentrations.
- B) Nitrogen absorbs all infrared radiation from Earth.
- C) Methane cannot interact with infrared radiation.
- D) Atmospheric gases have no effect on heat transfer.

GED-SCI11-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor records thyroid hormone level and thyroid-stimulating hormone (TSH) level in four patients.

Patient | Thyroid hormone | TSH level

A | Normal | Normal

B | Low | High

C | High | Low

D | Normal | Normal

In a typical negative feedback loop, low thyroid hormone can lead to higher TSH release.

Which patient's results best match low thyroid hormone causing increased TSH?

A) Patient A

B) Patient B

C) Patient C

D) Patient D

GED-SCI11-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

A) Angle at which the ramp is set

B) Time for the cart to travel 1 meter

C) Same cart used in every trial

D) Color of the ramp label

Targets:

T1) Independent variable

T2) Dependent variable

T3) Controlled variable

GED-SCI11-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The liver can convert excess glucose into glycogen for storage. Later, when blood glucose is low, the liver can break down glycogen and release glucose into the blood. This helps provide fuel for body cells between meals.

Which function of the liver is described?

- A) Helping regulate blood glucose storage and release
- B) Exchanging oxygen and carbon dioxide in air sacs
- C) Sending electrical impulses through neurons
- D) Producing eggs and sperm

GED-SCI11-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A roller coaster car is at the top of a hill. As it moves downward, its height decreases and its speed increases. Some gravitational potential energy changes into kinetic energy as the car moves.

Which energy change is mainly described?

- A) Gravitational potential energy changing into kinetic energy
- B) Electrical energy changing into nuclear energy
- C) Chemical energy changing into sound only
- D) Kinetic energy disappearing completely

GED-SCI11-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A town reduces its annual landfill waste from 8,000 tons to 6,400 tons after starting a recycling program.

What is the percent decrease in landfill waste?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI11-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a balance-training program reduces fall risk in older adults. Group A completes balance exercises for 8 weeks. Group B continues normal activity without the program. Both groups take the same balance test before and after the study.

Why is Group B included?

- A) It provides a comparison group for evaluating the training program.
- B) It guarantees that Group A will improve.
- C) It prevents researchers from collecting data.
- D) It changes normal activity into balance training.

GED-SCI11-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares how the mass of a swinging pendulum affects the time for one swing. The student tests 50 g, 100 g, and 150 g masses. The string length and release angle are kept the same.

What is the independent variable?

A) Mass of the pendulum bob

B) Time for one swing

C) String length

D) Release angle

GED-SCI11-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A compound machine is made from two or more simple machines. A wheelbarrow uses a wheel and axle plus a lever. The wheel and axle helps reduce [D1], while the handles act as a [D2].

Dropdowns:

D1) friction | DNA | precipitation

D2) lever | chromosome | glacier

GED-SCI11-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A physical therapist measures grip strength before and after a hand-strengthening program.

Participant group | Average grip strength before (kg) | Average grip strength after (kg)

Group A | 22 | 29

Group B | 21 | 22

Which claim is best supported by the data?

A) Group A showed a greater increase in average grip strength.

B) Group B showed the greater increase in average grip strength.

C) Both groups decreased in average grip strength.

D) Neither group changed at all.

GED-SCI11-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the input force needed to move the same heavy box using different simple machines.

Simple machine setup | Input force needed (N)

No machine | 300

Inclined plane | 180

Pulley system | 120

Lever system | 90

Which setup reduced the input force the most?

- A) No machine
- B) Inclined plane
- C) Pulley system
- D) Lever system

GED-SCI11-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A city compares two building materials for outdoor walkways.

Material | Sunlight reflected (%) | Surface temperature after 4 sunny hours (°C)

Dark brick | 12 | 49

Light stone | 46 | 34

Which TWO statements are best supported by the data?

- A) Light stone reflected a higher percentage of sunlight than dark brick.
- B) Dark brick reached a higher surface temperature than light stone.
- C) Dark brick reflected more sunlight than light stone.
- D) Both materials reached exactly the same temperature.
- E) Light stone had the higher surface temperature.

GED-SCI11-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A nerve impulse reaches a muscle cell. Calcium ions are released inside the muscle cell, allowing protein fibers to slide past each other. This sliding shortens the muscle cell and produces contraction.

Which process is mainly described?

- A) Muscle contraction
- B) Rock compaction
- C) Solar radiation
- D) Groundwater recharge

GED-SCI11-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student observes a skateboard rolling across a flat surface. The skateboard slows down and stops because friction and air resistance act opposite the direction of motion.

Which statement best explains why the skateboard stops?

- A) Unbalanced forces act opposite the skateboard's motion.
- B) Balanced forces make the skateboard speed up forever.
- C) Gravity stops existing on flat surfaces.
- D) Motion cannot change unless the object is in water.

GED-SCI11-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A geologist describes how limestone can form from the shells of marine organisms. Over long periods, shell fragments accumulate on the seafloor, become compacted, and are cemented together into rock.

Which rock type is most likely formed by this process?

- A) Sedimentary rock
- B) Igneous rock
- C) Metamorphic rock only
- D) Solid metal core material

GED-SCI11-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A clinic screens 280 adults for low vitamin D levels. Of these adults, 84 have low vitamin D levels.

What percent of the adults have low vitamin D levels?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI11-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student wants to test how surface texture affects how far a hockey puck slides. The student slides the same puck with the same starting push across smooth plastic, wood, and rough rubber.

Which factor should be measured as the dependent variable?

- A) Distance the puck slides
- B) Type of surface texture
- C) Starting push, kept the same
- D) Type of puck, kept the same

GED-SCI11-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A kidney nephron filters small molecules from blood. Useful substances such as glucose and much of the water are reabsorbed back into the blood, while extra wastes remain in the fluid that becomes urine.

Which statement best describes one function of nephrons?

- A) They filter blood and help form urine.
- B) They pump air into the lungs.
- C) They make sunlight into sugars.
- D) They move tectonic plates.

GED-SCI11-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the stopping distance of the same bicycle at different starting speeds on the same dry road.

Starting speed (m/s): 2, 4, 6, 8

Stopping distance (m): 1.5, 6.0, 13.5, 24.0

Which statement is best supported by the data?

- A) Stopping distance increased as starting speed increased.
- B) Stopping distance decreased as starting speed increased.
- C) Stopping distance stayed constant for all speeds.
- D) The bicycle stopped instantly at the highest speed.

GED-SCI11-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares four stars.

Star | Color | Surface temperature (K)

A | Red | 3,200

B | Yellow | 5,800

C | White | 9,500
D | Blue | 18,000

Which star has the highest surface temperature?

- A) Star A
- B) Star B
- C) Star C
- D) Star D

GED-SCI11-029

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

In a forest, some fungi grow in association with tree roots. The fungal network can help trees absorb minerals, while trees provide sugars to the fungi. In dry years, trees connected to these fungi may maintain better mineral uptake.

Which ecological relationship is described?

- A) Mutualism
- B) Predation
- C) Competition
- D) Parasitism only

GED-SCI11-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student compares two materials. Material A is easily shaped into thin sheets without breaking. Material B breaks into pieces when hammered. The ability to be shaped into sheets is called malleability.

Which statement is best supported?

- A) Material A is more malleable than Material B.
- B) Material B is more malleable than Material A.
- C) Both materials must be gases.
- D) Malleability means the same thing as magnetism.

GED-SCI11-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher tests whether a new ankle exercise improves balance after a sprain. Participants are assigned to an exercise group or a usual-care group. Balance is measured using the same test after four weeks.

Which TWO choices would improve the investigation?

- A) Use the same balance test for all participants.
- B) Include enough participants in both groups.
- C) Change the balance test for each participant.
- D) Record only results from participants who improved.
- E) Give only one group extra testing time.

GED-SCI11-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student measures the frequency and wavelength of several waves traveling at the same speed.

Wave | Frequency (Hz) | Wavelength (m)

- A | 5 | 12
- B | 10 | 6
- C | 15 | 4
- D | 20 | 3

Which wave has a speed of 60 m/s?

- A) Wave A only
- B) Waves A and B only
- C) Waves A, B, C, and D
- D) Wave D only

GED-SCI11-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A DNA sequence in a gene changes from GAA to GTA. This change leads to a different amino acid being placed in a protein. The altered protein does not bind its target molecule as well as the original protein.

Which statement best explains the result?

- A) A gene mutation can change a protein and affect how it works.
- B) DNA changes can never affect proteins.
- C) Proteins are made of planets and minerals.
- D) All gene mutations remove chromosomes completely.

GED-SCI11-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

After a severe drought, a pond becomes a meadow. Over time, grasses, shrubs, and then young trees

begin growing in the area. The community changes gradually as soil conditions and available resources change.

Which ecological process is described?

- A) Ecological succession
- B) Static electricity
- C) Boiling
- D) Chromosome separation

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI11-001 — Correct: A — Correct Answer: Sleep interruption was associated with lower recall scores. — Explanation: Group B had interrupted sleep and recalled fewer words on average than Group A.

GED-SCI11-002 — Correct: C — Correct Answer: Object C — Explanation: Potential energy = weight $\times$ height. Object C has $12\text{ N} \times 8\text{ m} = 96\text{ J}$, the greatest value listed.

GED-SCI11-003 — Correct: A — Correct Answer: It can reduce the amount of new copper ore that must be mined. — Explanation: Recycling copper can reduce demand for newly mined copper ore.

GED-SCI11-004 — Correct: A — Correct Answer: Hormones help regulate internal conditions through feedback. — Explanation: Insulin and glucagon help keep blood glucose within a normal range through feedback.

GED-SCI11-005 — Correct: 240 — Tolerance: exact — Correct Answer: 240 — Explanation: Work = $40\text{ N} \times 6\text{ m} = 240\text{ J}$.

GED-SCI11-006 — Correct: A,B — Correct Answer: Weight of the load | Type of lever used — Explanation: The load and lever should stay the same so lever-arm length is the main variable being tested.

GED-SCI11-007 — Correct: D1=codominance; D2=protein — Correct Answer: D1=codominance; D2=protein — Explanation: Both hemoglobin alleles can be expressed, and hemoglobin is a protein.

GED-SCI11-008 — Correct: A — Correct Answer: The liquid is acidic. — Explanation: pH 4 and bubbling with baking soda are consistent with an acidic liquid.

GED-SCI11-009 — Correct: A — Correct Answer: Methane can influence atmospheric warming even at lower concentrations. — Explanation: Methane absorbs infrared radiation strongly, so it can affect heat trapping even in smaller amounts.

GED-SCI11-010 — Correct: B — Correct Answer: Patient B — Explanation: Patient B has low thyroid hormone and high TSH, matching the described negative feedback response.

GED-SCI11-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Angle at which the ramp is set; T2=Time for the cart to travel 1 meter; T3=Same cart used in every trial — Explanation: Ramp angle is changed, travel time is measured, and the same cart is controlled.

GED-SCI11-012 — Correct: A — Correct Answer: Helping regulate blood glucose storage and release — Explanation: The liver stores glucose as glycogen and releases glucose when needed.

GED-SCI11-013 — Correct: A — Correct Answer: Gravitational potential energy changing into kinetic energy — Explanation: As height decreases and speed increases, gravitational potential energy changes into kinetic energy.

GED-SCI11-014 — Correct: 20 — Tolerance: exact — Correct Answer: 20 — Explanation: Waste decreased by 1,600 tons. $1,600 \div 8,000 \times 100 = 20\%$.

GED-SCI11-015 — Correct: A — Correct Answer: It provides a comparison group for evaluating the training program. — Explanation: The comparison group helps show whether changes are related to the training program.

GED-SCI11-016 — Correct: A — Correct Answer: Mass of the pendulum bob — Explanation: The mass is the factor intentionally changed in the experiment.

GED-SCI11-017 — Correct: D1=friction; D2=lever — Correct Answer: D1=friction; D2=lever — Explanation: A wheel and axle can reduce friction, and wheelbarrow handles act as a lever.

GED-SCI11-018 — Correct: A — Correct Answer: Group A showed a greater increase in average grip strength. — Explanation: Group A increased by 7 kg, while Group B increased by 1 kg.

GED-SCI11-019 — Correct: D — Correct Answer: Lever system — Explanation: The lever system required the least input force, 90 N, so it reduced input force the most.

GED-SCI11-020 — Correct: A,B — Correct Answer: Light stone reflected a higher percentage of sunlight than dark brick. | Dark brick reached a higher surface temperature than light stone. — Explanation: Light stone reflected 46% and reached 34°C, while dark brick reflected 12% and reached 49°C.

GED-SCI11-021 — Correct: A — Correct Answer: Muscle contraction — Explanation: Sliding protein fibers shorten muscle cells, producing muscle contraction.

GED-SCI11-022 — Correct: A — Correct Answer: Unbalanced forces act opposite the skateboard's motion. — Explanation: Friction and air resistance oppose motion and slow the skateboard.

GED-SCI11-023 — Correct: A — Correct Answer: Sedimentary rock — Explanation: Compacted and cemented shell fragments can form sedimentary limestone.

GED-SCI11-024 — Correct: 30 — Tolerance: exact — Correct Answer: 30 — Explanation: $84 \div 280 \times 100 = 30\%$.

GED-SCI11-025 — Correct: A — Correct Answer: Distance the puck slides — Explanation: The dependent variable is the measured result after changing surface texture.

GED-SCI11-026 — Correct: A — Correct Answer: They filter blood and help form urine. — Explanation: Nephrons filter blood, reabsorb useful substances, and help form urine.

GED-SCI11-027 — Correct: A — Correct Answer: Stopping distance increased as starting speed increased. — Explanation: The stopping distance rose from 1.5 m to 24.0 m as starting speed increased.

GED-SCI11-028 — Correct: D — Correct Answer: Star D — Explanation: Star D has the highest listed surface temperature, 18,000 K.

GED-SCI11-029 — Correct: A — Correct Answer: Mutualism — Explanation: Both fungi and trees benefit from the relationship.

GED-SCI11-030 — Correct: A — Correct Answer: Material A is more malleable than Material B. — Explanation: Material A can be shaped into sheets, while Material B breaks when hammered.

GED-SCI11-031 — Correct: A,B — Correct Answer: Use the same balance test for all participants. | Include enough participants in both groups. — Explanation: A consistent test and adequate sample size improve reliability.

GED-SCI11-032 — Correct: C — Correct Answer: Waves A, B, C, and D — Explanation: Wave speed = frequency $\times$ wavelength. Each wave has speed 60 m/s.

GED-SCI11-033 — Correct: A — Correct Answer: A gene mutation can change a protein and affect how it works. — Explanation: A DNA sequence change can alter an amino acid and affect protein function.

GED-SCI11-034 — Correct: A — Correct Answer: Ecological succession — Explanation: Succession is the gradual change in a biological community over time.